K-Nearest Neighbours for Breast Cancer Diagnosis

Implementation, optimisation and overfitting analysis

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Missing a cancer costs more than a false alarm

False alarm

A benign tumour is flagged. The patient has further tests.

Missed cancer

A malignant tumour is called benign. Treatment is delayed.

Our three questions

1. How accurately can K-NN separate malignant from benign tumours?
2. Which optimisation techniques actually help?
3. Where does the model overfit or underfit?

The data: 569 tumours, 30 measurements

569
tumour samples

357
benign

212
malignant

30
features per sample, such as radius and texture

455
training samples (80%)

114
test samples (20%), held out

Features are standardised using training statistics only, which prevents data leakage. Source: UCI Machine Learning Repository, also on Kaggle.

Our pipeline in six steps

K-NN finds the k closest training samples and predicts their majority class.

1 Split

80/20 stratified, test set held out

2 Standardise

Training statistics only

3 Compare metrics

Euclidean, Manhattan, cosine, Mahalanobis

4 Tune

k from 1 to 25, plus PCA

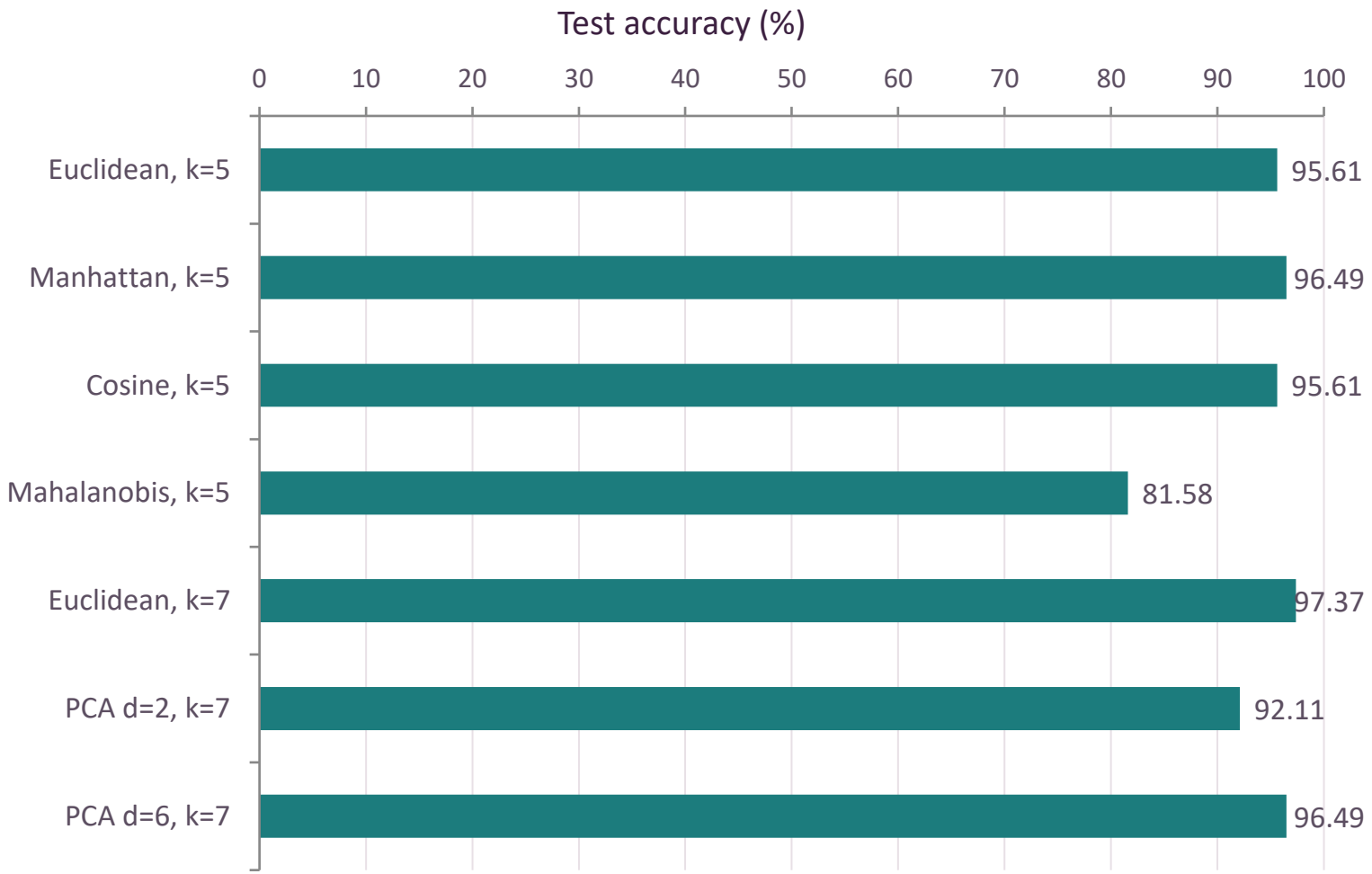
5 Optimise

Soft-KNN: optimisers, activations, augmentation

6 Calibrate

Set the vote threshold, then evaluate

Most configurations reach about 95 to 97%



96.55%

K-NN, k=7, repeated cross-validation (± 1.61). Our fairest estimate.

0.88 pts

The accuracy change from one test sample. Small gaps can be noise.

Optimisation rarely beat the baseline

Technique	Setting	Test accuracy	Malignant recall
Optimiser	AdamW	95.53% ± 0.92	96.43%
	SGD + momentum	95.18% ± 1.13	95.24%
	SGD, no momentum	94.04% ± 1.75	93.57%
Activation	ReLU (AdamW)	95.18% ± 0.71	95.71%
Regularisation	Early stopping, epoch 9	95.26% ± 1.05	95.00%
Augmentation	K-NN + jitter, std 0.3	96.23% ± 0.79	92.86%

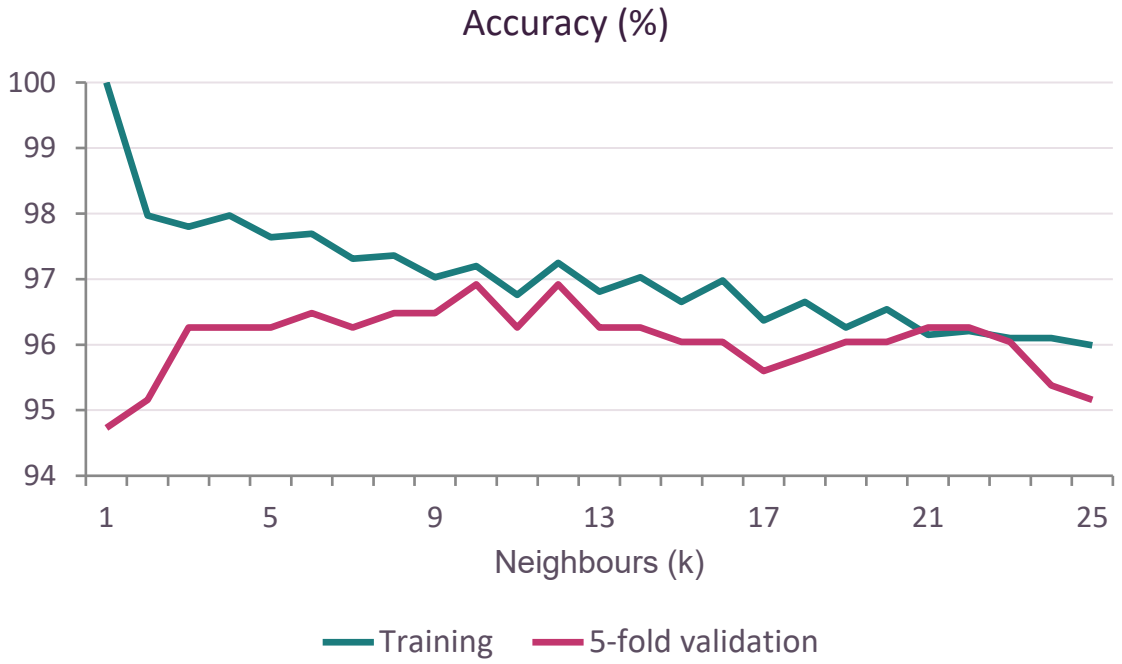
Averaged over 10 seeds

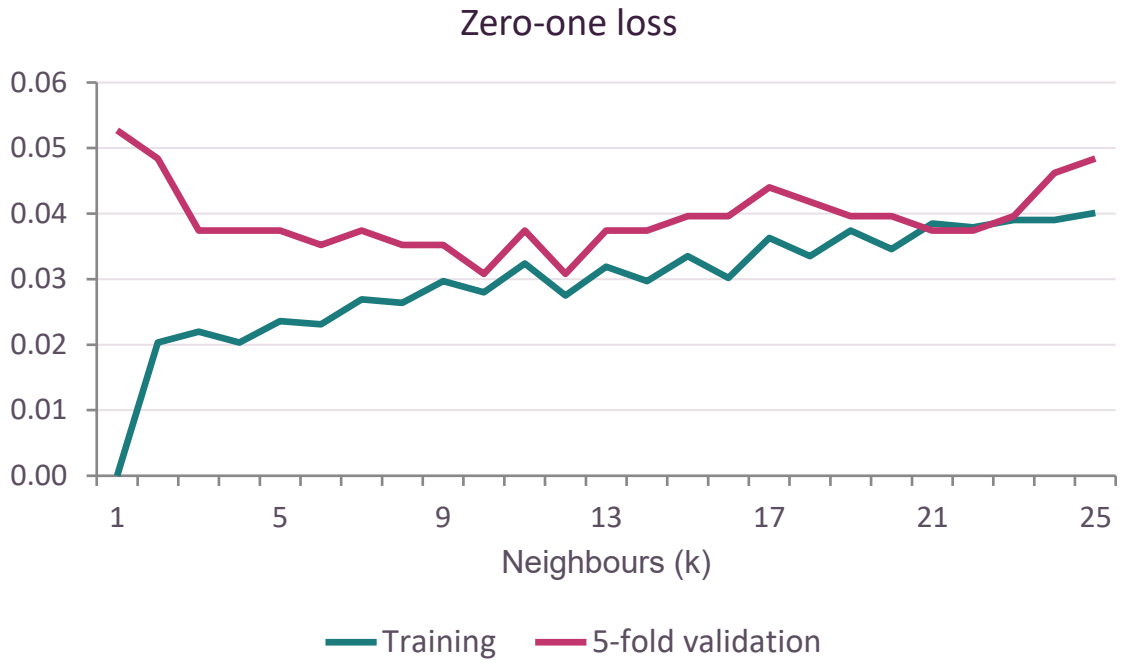
AdamW runs ranged from 93.86% to 96.49%, depending on the seed.

About 22× faster code

NumPy broadcasting returned identical predictions to Python loops.

K-NN: k=1 overfits, k=25 starts to underfit





k = 1: overfitting

Train 100.00%, validation 94.73%

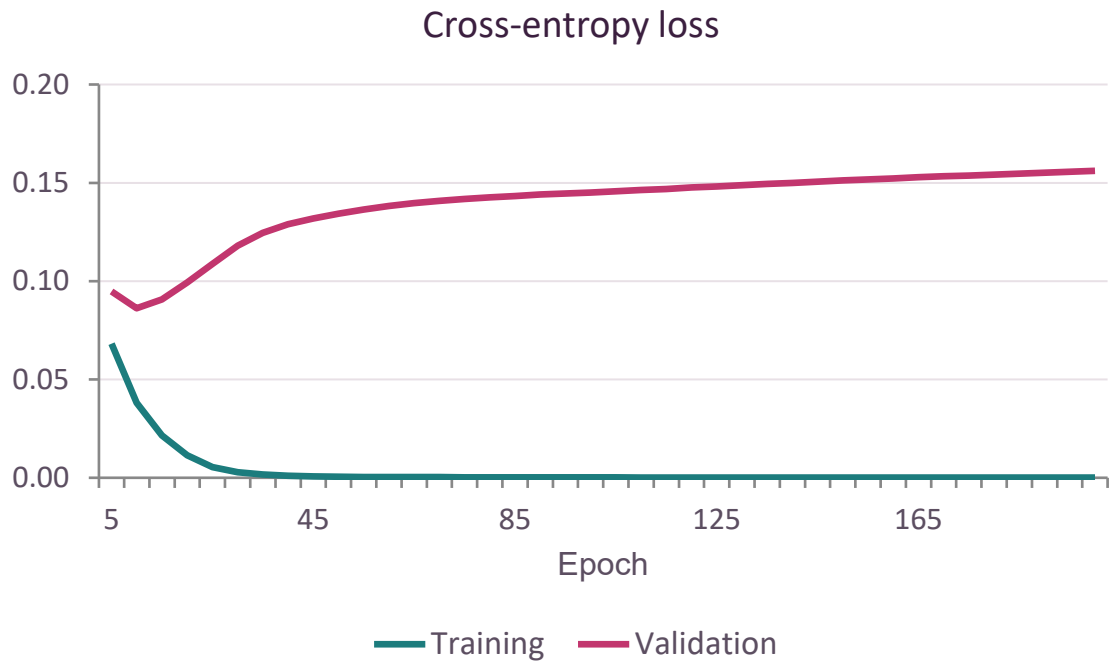
k = 7: balanced

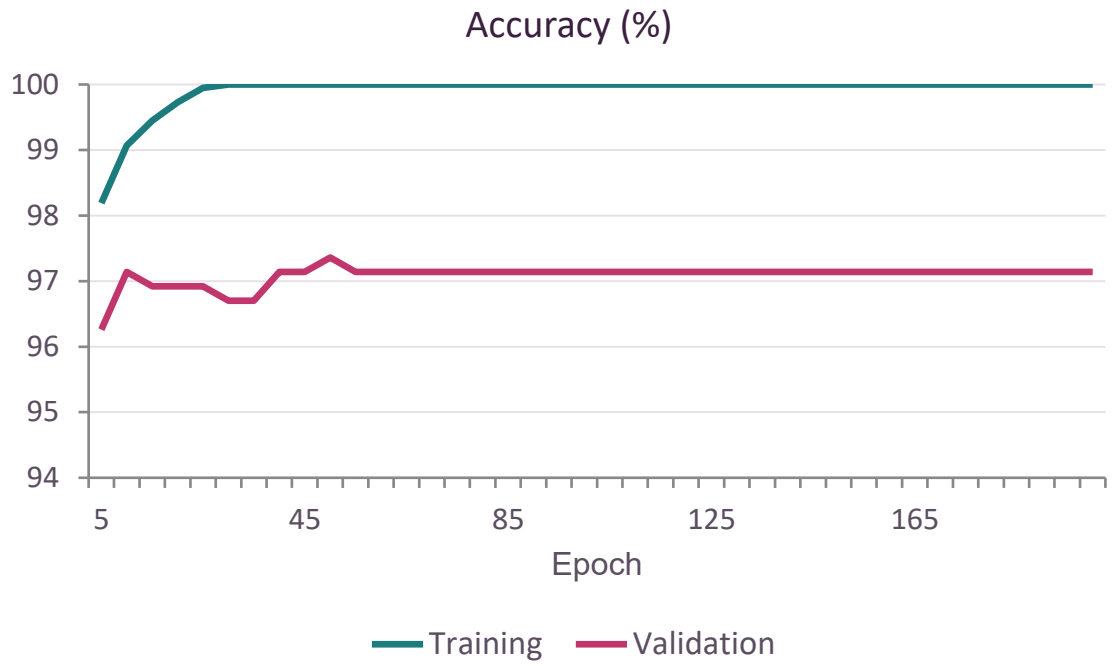
Train 97.31%, validation 96.26%

k = 25: mild underfitting

Train 95.99%, validation 95.16%

Soft-KNN: loss reveals overfitting that accuracy hides





0.086

lowest validation loss, at epoch 9

0.156

validation loss at epoch 200

≈ 0

training loss at epoch 200

A lower vote threshold catches one more cancer

Default: 4 of 7 votes

Predicted benign / malignant

72 true benign	**0** false alarms
3 missed	**39** cancers caught

Safety: 3 of 7 votes

Predicted benign / malignant

66 true benign	**6** false alarms
2 missed	**40** cancers caught

92.86% → 95.24%

Malignant recall improves, while accuracy falls from 97.37% to 92.98%.

Test patient #35

Malignant, with 3 of 7 malignant votes. Only the safety rule catches it.

Illustrative: examined on the test split.

Checking our results changed some headline claims

Earlier claim	Verified result
97.37% accuracy with 97.6% recall	No single model achieves both
Zero-miss clinical recall	2 cancers still missed at 3 of 7
Soft-KNN (AdamW) 96.49%	95.53% ± 0.92 over 10 seeds
122× speedup	About 22× on our machine
Convex fusion model, 97.37%	Excluded: no code in the repository

We recommend K-NN with k=7

Recommended

Standardised K-NN with k=7. Expected accuracy 96.55% ± 1.61. Simple, explainable and low overfitting risk.

When recall matters most

The 3-of-7 rule (2 missed, 6 false alarms) or the Soft-KNN ensemble (1 missed, 4 false alarms).

Limitations

Small, single-source data. The test split informed several comparisons. Not a medical device.

Next steps

Nested cross-validation for the threshold. External validation on data from another hospital.

Thank you

Simple K-NN performs strongly, and thresholds matter more than tuning.

Code, results and full report:
github.com/Swayam424/breast-cancer-knn-optimization